<u>Hyperfine coupling</u> (Coupling between nuclear spin and electron spin for electron in $l=0$ state)

Proton $\vec{\mu}_p = \frac{g_p e}{2m_p}\vec{S}_p$, $\vec{\mu}_e = -\frac{e}{m}\vec{S}_e$

$g_p = 5.59$ $\qquad g_e = 2.00$

Magnetic field of a dipole $\vec{B} = \frac{\mu_0}{4\pi r^3}\left(3(\vec{\mu}\cdot\hat{r})\hat{r} - \vec{\mu}\right) + \frac{2\mu_0}{3}\vec{\mu}\,\delta^3(\vec{r})$

(can be derived by considering spinning charged sphere)

$$H'_{hf} = -\vec{\mu}_e\vec{B}_p = \frac{\mu_0 g_p e^2}{8\pi m_p m_e}\frac{3(\vec{S}_p\cdot\hat{r})(\vec{S}_e\cdot\hat{r}) - \vec{S}_p\cdot\vec{S}_e}{r^3}$$

$$+ \frac{\mu_0 g_p e^2}{3 m_p m_e}\vec{S}_p\cdot\vec{S}_e\,\delta^3(\vec{r})$$

$\Rightarrow$ 1st order perturbative correction is just (average over electron's spatial w.f.)

$$E^1_{hf} = \frac{\mu_0 g_p e^2}{8\pi m_p m_e}\left\langle\frac{3(\vec{S}_p\cdot\hat{r})(\vec{S}_e\cdot\hat{r}) - \vec{S}_p\cdot\vec{S}_e}{r^3}\right\rangle + \frac{\mu_0 g_p e^2}{3 m_p m_e}|\psi(0)|^2\langle\vec{S}_p\cdot\vec{S}_e\rangle$$

electron's w.f. at the nuclei

P G 6.27 $J = \int(\vec{a}\cdot\hat{r})(\vec{b}\cdot\hat{r})\sin\theta\,d\theta\,d\varphi = \frac{4\pi}{3}(\vec{a}\cdot\vec{b})$

Using $\hat{r} = \sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$

$$J = \int\left(a_x\sin\theta\cos\varphi + a_y\sin\theta\sin\varphi + a_z\cos\theta\right)\left(b_x\sin\theta\cos\varphi + b_y\sin\theta\sin\varphi + b_z\cos\theta\right)\sin\theta\,d\theta\,d\varphi$$

$=$ (need only $\cos^2\varphi$, $\sin^2\varphi$)

$$= \int_0^{2\pi}d\varphi\int_{-1}^{1}d(\cos\theta)\left[\left(a_x b_x\cos^2\varphi + a_y b_y\sin^2\varphi\right)\sin^2\theta + a_z b_z\cos^2\theta\right]$$

$$= 2\pi\cdot\frac{2}{3}a_z b_z + \pi(a_x b_x + a_y b_y)\underbrace{2\int_0^1 dx\,(1-x^2)}_{2\cdot\left(1-\frac{1}{3}\right)=\frac{4}{3}} = \frac{4\pi}{3}\vec{a}\cdot\vec{b}$$

$$\Rightarrow \left\langle \frac{3(\vec{S}_p\cdot\hat{r})(\vec{S}_e\cdot\hat{r}) - \vec{S}_p\cdot\vec{S}_e}{r^3} \right\rangle =$$

$$= \int_0^\infty dr\, r^2 \frac{1}{r^3} R^2(r) \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \left[3(S_p\cdot\hat{r})(S_e\cdot\hat{r}) - S_p\cdot S_e\right]$$

$$= \int_0^\infty \frac{dr}{r} R^2(r) \left[\frac{4\pi}{3}(\vec{S}_p\cdot\vec{S}_e)\cdot 3 - (\vec{S}_p\cdot\vec{S}_e)\,4\pi\right] = 0$$

as long as angular w.f. is constant $\rightarrow$ $l=0$ state.

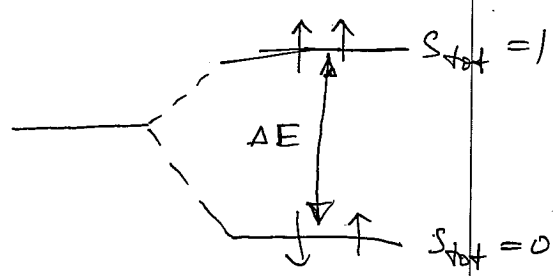
$$\Rightarrow E^1_{hf} = \frac{\mu_0 g_p e^2}{3 m_p m_e} \frac{1}{\pi a^3} \frac{1}{2} \left\langle \left(\vec{S}^2_{tot} - \vec{S}^2_e - \vec{S}^2_p\right)\right\rangle$$

$$\vec{S}^2_{tot} = (\vec{S}_p + \vec{S}_e)^2 = \hbar^2 S_{tot}(S_{tot}+1) = \hbar^2 \begin{cases} 2, & S_{tot}=1 \quad \text{triplet} \\ 0, & S_{tot}=0 \quad \text{singlet} \end{cases}$$

$$\vec{S}^2_e = \hbar^2 \frac{1}{2}\left(\frac{1}{2}+1\right) = \frac{3}{4}\hbar^2$$

$$\Rightarrow \frac{1}{2}\langle\ \rangle = \frac{\hbar^2}{2} \begin{cases} 2 - \frac{3}{2} & \text{triplet} \\ 0 - \frac{3}{2} & \text{singlet} \end{cases} = \begin{cases} \frac{\hbar^2}{4}, & \text{triplet} \\ -\frac{3}{4}\hbar^2, & \text{singlet} \end{cases}$$

$$E^1_{hf} = \frac{\mu_0 g_p e^2 \hbar^2}{3\pi m_p m_e a^3} \cdot \begin{cases} \frac{1}{4} \\ -\frac{3}{4} \end{cases}$$



But $\mu_0\varepsilon_0 = \frac{1}{c^2} \Rightarrow \mu_0 = \frac{1}{\varepsilon_0 c^2}$

$$\Delta E_{hf} = \frac{g_p e^2 \hbar^2}{3\pi m_p m_e \varepsilon_0 c^2 a^3} = \frac{4 g_p \hbar^4}{3 m_p m_e^2} \underbrace{\left(\frac{e^2 m_e}{4\pi\varepsilon_0 \hbar^2}\right)}_{=\frac{1}{a}} \frac{1}{c^2 a^3} =$$

$$= \frac{4 g_p \hbar^4}{3 m_p m_e^2 c^2 a^4} = 5.88\cdot 10^{-6}\,eV$$

$f = \frac{\Delta E}{h} = 1420\ MHz$, $\lambda = \frac{c}{f} = 21\ cm$ (microwave range) & 3K radiation

$$E_1 = -\frac{\hbar^2}{2 m_e a^2}$$

$$\Delta E_{hf} = \frac{4 g_p}{3 m_p c^2} \left(\frac{\hbar^2}{2 m_e a^2}\right)^2 \cdot 4 = \frac{16}{3} g_p \frac{(E_1)^2}{m_p c^2} =$$

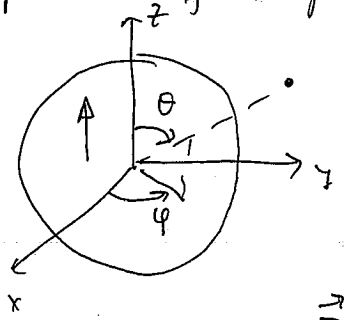
$$= \frac{16}{3} (5.59) \frac{(13.6\,eV)^2}{1.67 \cdot 10^{-27}\,kg \cdot \left(3 \cdot 10^8 \frac{m}{s}\right)^2} = \frac{16}{3} \frac{5.6 \cdot 185}{93.8 \cdot 10^7}\,eV = 5.89 \cdot 10^{-6}\,eV$$

$$1.67 \cdot 9 \cdot 10^{-11}\,J = 1.67 \cdot 9 \cdot 10^{-11}\; 6.24 \cdot 10^{18}\,eV$$

Explanation of the contact field $\frac{2\mu_0}{3}\vec{\mu}\,\delta^3(\vec{r})$

in $\vec{B}_{dip} = \frac{\mu_0}{4\pi r^3}\left(3(\vec{\mu}\cdot\hat{r})\hat{r} - \vec{\mu}\right) + \underline{\frac{2\mu_0}{3}\vec{\mu}\,\delta^3(\vec{r})}$

Think of dipole as of uniformly magnetized sphere:



$\vec{M} = M\hat{z}$ inside sphere

$\vec{B} = \mu_0(\vec{H} + \vec{M})$

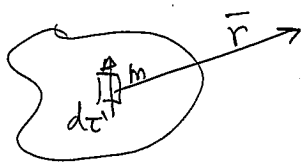
$\vec{\nabla}\cdot\vec{B} = 0$ and $\vec{M} = 0$ outside

$\Rightarrow \vec{\nabla}\cdot\vec{H} = -\vec{\nabla}\cdot\vec{M}$ on the surface (demagnetizing field)

Start with vector potential of single dipole

$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\hat{r}}{r^2}$ $\quad\left(\vec{B} = \vec{\nabla}\times\vec{A} \Rightarrow \vec{\nabla}\cdot\vec{B} = 0 \text{ always}\right)$

distribution:



$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{m}(r')\times\hat{r}}{\mathscr{r}^2}\,d\tau'$$

$$\nabla'\frac{1}{\mathscr{r}} = \frac{\hat{\mathscr{r}}}{\mathscr{r}^2}$$

$$\Rightarrow \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int\left[\vec{m}(\vec{r}\,') \times \nabla'\frac{1}{\mathscr{r}}\right]d\tau' = \left(\text{Product rule 7 of Griffiths}\right) =$$

$$= \frac{\mu_0}{4\pi}\int \frac{1}{\mathscr{r}}\,\nabla'\times\vec{M}(\vec{r}\,')\,d\tau' - \frac{\mu_0}{4\pi}\int \nabla'\times\left(\frac{\vec{M}(r')}{\mathscr{r}}\right)d\tau'.$$

Now, for any $\vec{v}$: $\int_V \vec{\nabla}\cdot\vec{v}\,d\tau = \int_S \vec{v}\cdot d\vec{a}$

let $\vec{v} \to \vec{v}\times\vec{c}$ where $\vec{c}$ = constant $\Rightarrow \vec{\nabla}\cdot(\vec{v}\times\vec{c}) = \vec{c}\cdot(\vec{\nabla}\times\vec{v}) - \vec{v}\cdot\underbrace{(\vec{\nabla}\times\vec{c})}_{0}$

$$\Rightarrow \int_V \vec{c}\cdot(\vec{\nabla}\times\vec{v})\,d\tau = \int_S \vec{v}\times\vec{c}\cdot d\vec{a} = \int_S \vec{c}\cdot(-\vec{v}\times d\vec{a})$$

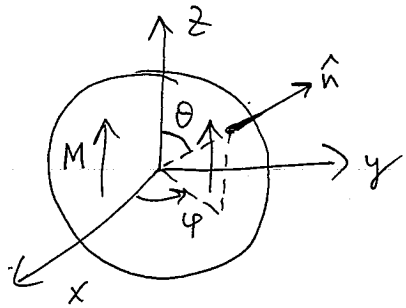
$d\vec{a}\cdot\vec{v}\times\vec{c} = \vec{c}\cdot d\vec{a}\times\vec{v}$

$$\Rightarrow \int_V \vec{\nabla}\times\vec{v}\,d\tau = -\int_S \vec{v}\times d\vec{a}$$

Thus $\vec{A}(\bar{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\bar{\nabla}' \times \bar{M}(r')}{\mathscr{r}} d\bar{\tau}' + \frac{\mu_0}{4\pi} \oint \frac{\vec{M}(r') \times d\vec{a}'}{\mathscr{r}}$

Volume current $\vec{J}_b = \vec{\nabla} \times \vec{M}$

Surface $\vec{K}_b = \vec{M} \times \hat{n}$ $\quad d\vec{a} = da\, \hat{n}$

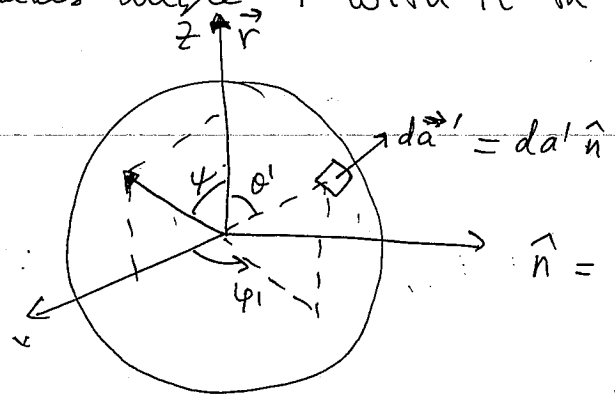


$\vec{J}_b = 0$ for $\vec{M} = \text{const}$

but $\vec{K}_b = \vec{M} \times \vec{n} = M\hat{z} \times \hat{n} = M \sin\theta\, \hat{\varphi}$

Choose observation point $\vec{r}$ along $\hat{z}$ axis.

$\vec{M}$ makes angle ψ with it in xz plane: $\vec{M} = M(\sin\psi\,\hat{x} + \cos\psi\,\hat{z})$



$$\hat{n} = \sin\theta'\cos\varphi'\,\hat{x} + \sin\theta'\sin\varphi'\,\hat{y} + \cos\theta'\,\hat{z}$$

$$\vec{A}(r) = \frac{\mu_0}{4\pi}\oint \frac{\vec{M}(r')\times\hat{n}}{|\vec{r}-\vec{r}'|}\,da' \quad , \quad da' = R^2\sin\theta'\,d\theta'\,d\varphi'$$

$$|\vec{r}-\vec{r}'| = \sqrt{R^2 + r^2 - 2Rr\cos\theta'}$$

$$\vec{M}(r')\times\hat{n} = \begin{vmatrix} x & y & z \\ \sin\psi & 0 & \cos\psi \\ \sin\theta'\cos\varphi' & \sin\theta'\sin\varphi' & \cos\theta' \end{vmatrix} = \hat{x}\left(-\cos\psi\sin\theta'\sin\varphi'\right)$$

$$-\hat{y}\left(\sin\psi\cos\theta' - \cos\psi\sin\theta'\cos\varphi'\right) + \hat{z}\left(\sin\psi\sin\theta'\sin\varphi'\right)$$

$\Rightarrow$ Integration over $d\varphi'$ kills all but $(-\hat{y}\sin\psi\cos\theta')$.

$$\vec{A}(r) = \frac{\mu_0 M}{4\pi}\cdot 2\pi\left(-\hat{y}\sin\psi\right)R^2\int_0^{\pi} d\theta'\,\frac{\sin\theta'\cos\theta'}{\sqrt{r^2+R^2-2rR\cdot\cos\theta'}}$$

let $\cos\theta' = u \Rightarrow$ $\displaystyle\int_{-1}^{1}\frac{+du\cdot u}{\sqrt{r^2+R^2-2rRu}} = \left(u = \frac{x}{2Rr}\right) =$

$$= \frac{1}{4R^2r^2}\int_{-2Rr}^{2Rr}\frac{dx\;x}{\sqrt{r^2+R^2-x}} = \frac{-1}{4R^2r^2}\int dx\,\frac{(a-x-a)}{\sqrt{a-x}} =$$

$$= \frac{-1}{(2Rr)^2}\left[2a\sqrt{a-x} - \frac{2}{3}(a-x)^{3/2}\right] = \frac{1}{(2Rr)^2}\,\frac{2}{3}\sqrt{a-x}\left[a-x-3a\right] =$$

$$= \frac{1}{(2Rr)^2}\,\frac{-2}{3}\left(2Rru + 2R^2 + 2r^2\right)\sqrt{R^2+r^2-2Rr}\,\Big|_{-1}^{1} = \frac{-1}{3R^2r^2}\Big\{(R^2+r^2+Rr)|R-r|$$

$$-(R^2+r^2-Rr)(R+r)\Big\}$$

$$\Rightarrow \vec{A}(\vec{r}) = (+\hat{y}\sin\psi)\,\frac{\mu_0 M}{2}\,\frac{1}{3r^2}\left\{(R^2+r^2+Rr)\,|R-r| - (R^2+r^2-Rr)\,|R+r|\right\}$$

$\underline{r<R} \Rightarrow |R-r| = R-r$

$$\frac{1}{r^2}\left\{(R^2+r^2+Rr)(R-r) - (R^2+r^2-Rr)(R+r)\right\} =$$

$$= \frac{1}{r^2}\left\{-2r(R^2+r^2) + R^2 r + R^2 r\right\} = -2r$$

$\underline{r>R} \Rightarrow |R-r| = r-R$

$$\frac{1}{r^2}\left\{(R^2+r^2+Rr)(r-R) - (R^2+r^2-Rr)(R+r)\right\} =$$

$$= \frac{1}{r^2}\left\{-2R(R^2+r^2) + 2Rr^2\right\} = -\frac{2R^3}{r^2}$$

$$\Rightarrow \vec{A}(\vec{r}) = (-\hat{y}\sin\psi)\,\frac{\mu_0 M}{3}\begin{cases} r & r<R \\ \dfrac{R^3}{r^2} & r>R \end{cases}$$

Observe now: $-\hat{y}\sin\psi = \dfrac{\vec{M}}{M}\times\dfrac{\vec{r}}{r}$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{3}\,\vec{M}\times\vec{r}\begin{cases} 1 & , r<R \\ \dfrac{R^3}{r^3} & , r>R \end{cases}$$

Thus, at long distances $(r>R)$ $\vec{A}(\vec{r}) = \dfrac{\mu_0}{3}\,\dfrac{\vec{M}R^3\times\hat{r}}{r^2}$

describes magnetic dipole with moment $\boxed{\vec{m}_d = \frac{4\pi}{3}R^3\vec{M}}$

Inside the sphere $(r<R)$: $\vec{A} = \dfrac{\mu_0}{3}(-\hat{x}\,M_y + \hat{y}\,M_x)$

$$\Rightarrow \vec{B} = \vec{\nabla}\times\vec{A} = \frac{\mu_0}{3}\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ -M_y & M_x & 0 \end{vmatrix} = \frac{\mu_0}{3}\,\hat{z}(2M) = \frac{2\mu_0}{3}\vec{M} = \frac{\mu_0}{2\pi}\,\frac{\vec{m}_d}{R^3}$$

constant magnetic field

As $R\to 0$ $\displaystyle\int_V \vec{B}\,d\tau = \frac{\mu_0}{2\pi}\,\frac{\vec{m}_d}{R^3}\cdot\frac{4\pi}{3}R^3 = \frac{2\mu_0}{3}\vec{m}_d$ reduces to constant!

$\Rightarrow \vec{B}(r<R) = \dfrac{2\mu_0}{3}\vec{m}_d\,\delta^3(\vec{r})$. This is the origin of the last term.